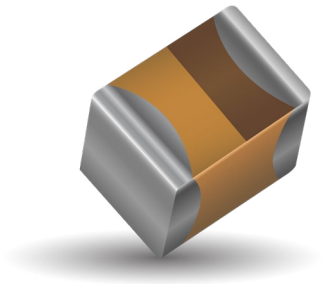
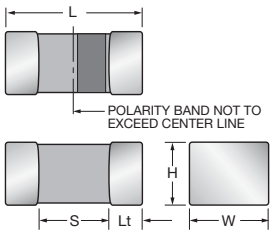
TAC Series

Standard and Low Profile Tantalum Microchip Capacitors





FEATURES

- The World's Smallest Surface Mount Tantalum Capacitor
- 100% Surge Current Tested
- CV Range: 0.10-150µF / 2-25V
- 10 Case Sizes Available, Standard and Low Profile

LEAD-FREE

LEAD-FREE COMPATIBLE COMPONENT

RoHS COMPLIANT

APPLICATIONS

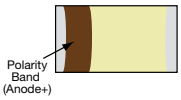
- Hearing Aids, Non-Life Support Medical, Long Life Miniature Designs
- Industrial and Hand-held and Wearable Applications

STANDARD CASE DIMENSIONS: millimeters (inches)

Code	EIA Code	EIA Metric	L+0.20 (0.008) -0.00 (0.000)	W+0.15 (0.006) -0.00 (0.000)	H+0.15 (0.006) -0.00 (0.000)	Termination Spacing(S)	Minimum Termination Length (Lt)
A	1206	3216-18	3.20 ± 0.20 (0.126 ± 0.008)	1.60 ± 0.20 (0.063 ± 0.008)	1.60 ± 0.20 (0.063 ± 0.008)	1.80 (0.071) min	0.15 (0.006)
B	1210	3528-15	3.50 ± 0.20 (0.138 ± 0.008)	$2.80^{+0.20}_{-0.10}$ ($0.110^{+0.008}_{-0.004}$)	1.50 (0.059) max	2.00 (0.079) min	0.15 (0.006)
K	0402	1005-07	1.00 (0.039)	$0.50^{+0.20}_{-0.00}$ ($0.020^{+0.008}_{-0.000}$)	$0.50^{+0.20}_{-0.00}$ ($0.020^{+0.008}_{-0.000}$)	0.40 (0.016) min	0.10 (0.004)
L	0603	1608-10	1.60 (0.063)	0.85 (0.033)	0.85 (0.033)	0.55 (0.022) min	0.15 (0.006)
R	0805	2012-15	2.00 (0.079)	1.35 (0.053)	1.35 (0.053)	0.70 (0.028) min	0.15 (0.006)

MARKING

A, B, H, I, K, L, R, T, U, V CASE



LOW PROFILE CASE DIMENSIONS: millimeters (inches)

Code	EIA Code	EIA Metric	L+0.20 (0.008) -0.00 (0.000)	W+0.15 (0.006) -0.00 (0.000)	H max	Termination Spacing(S)	Minimum Termination Length (Lt)
H	0805	2012-10	2.00 (0.079)	1.35 (0.053)	1.00 (0.039)	0.70 (0.028) min	0.15 (0.006)
I	1206	3216-05	3.20±0.20 (0.126±0.008)	1.60±0.20 (0.063±0.008)	0.50 (0.020)	1.80 (0.071) min.	0.15 (0.006)
T	1210	3528-12	3.50 ± 0.20 (0.138 ± 0.008)	$2.80^{+0.20}_{-0.10}$ ($0.110^{+0.008}_{-0.004}$)	1.20 (0.047)	2.00 (0.079) min	0.15 (0.006)
U	0805	2012-06	2.00 (0.079)	1.35 (0.053)	0.60 (0.024)	0.70 (0.028) min	0.15 (0.006)
V	1206	3216-08	3.20 ± 0.20 (0.126 ±0.008)	1.60 ± 0.20 (0.063 ± 0.008)	0.75 (0.030)	1.80 (0.071) min	0.15 (0.006)

HOW TO ORDER

TAC	L	226	*	004	R	TA
Type TACmicrochip®	**Case Size** See table above	**Capacitance Code** pF code: 1st two digits represent significant figures, 3rd digit represents multiplier (number of zeros to follow)	**Tolerance** K = ±10% M = ±20%	**Rated DC Voltage** 002=2Vdc 003=3Vdc 004=4Vdc 006=6.3Vdc 010=10Vdc 016=16Vdc 020=20Vdc 025=25Vdc	**Packaging** R, P = 7" Standard Tin Termination Plastic Tape X, Q = 41⁄4" Standard Tin Termination Plastic Tape A, M = 7" Gold Termination Plastic Tape F, N = 4¼" Gold Termination Plastic Tape	**Alternative characters may be used for special requirements**

TECHNICAL SPECIFICATIONS

Technical Data:	All technical data relate to an ambient temperature of +25°C								
Capacitance Range:	0.10 µF to 150 µF								
Capacitance Tolerance:	±10%; ±20%								
Leakage Current DCL:	0.01CV or 0.5µA whichever is the greater								
Rated Voltage (V_R) ≤ +85°C:	2	3	4	6.3	10	16	20	25	
Category Voltage (V_C) ≤ +125°C:	1.3	2	2.7	4	7	10	13	17	
Surge Voltage (V_S) ≤ +85°C:	2.7	3.9	5.2	8	13	20	26	32	
Surge Voltage (V_S) ≤ +125°C:	1.7	2.6	3.2	5	8	12	16	20	
Temperature Range:	-55°C to +125°C								
Reliability:	1% per 1000 hours at 85°C, V_R with 0.1Ω/V series impedance, 60% confidence level								
Termination Finish:	Tin Plating over Nickel (standard), Gold Plating over Nickel option available upon request								

The Important Information/Disclaimer is incorporated in the catalog where these specifications came from or available online at www.kyocera-avx.com/disclaimer/ by reference and should be reviewed in full before placing any order.

STANDARD MICROCHIP CAPACITANCE AND RATED VOLTAGE RANGE (LETTER DENOTES CASE SIZE)

Capacitance		Voltage Rating DC (V_R) at 85°C							
µF	Code	2.0V	3.0V	4.0V	6.3V	10V	16V	20V	25V
0.10	104						K		
0.15	154					K	K		
0.22	224					K	K	K	
0.33	334					K	K		
0.47	474					K/L	L		
0.68	684					K/L	L		
1.0	105				K/L	K/L/R	L		R
1.5	155			L	L	L			
2.2	225		K/L	L	K/L	L	L		
3.3	335	K/L	K/L	L	L	L/R		R	
4.7	475	K/L	K/L	L	L	L/R		R	
6.8	685	K/L	L	L	L/R	L/R			
10	106	K/L	L	L/R	L/R	L/R	R		
15	156		R	L/R	L/R	R			
22	226	R	L/R	L/R	R	R			
33	336	R	R	R	R	A/R			
47	476	R	R	R	A/R	B			
68	686	R	A/R	A					
100	107		A/R	A/R	A				
150	157	A							
220	227								

LOW PROFILE MICROCHIP CAPACITANCE AND RATED VOLTAGE RANGE (LETTER DENOTES CASE SIZE)

Capacitance		Voltage Rating DC (V_R) at 85°C					
µF	Code	2.0V	3.0V	4.0V	6.3V	10V	16V
1.0	105						U
1.5	155						
2.2	225					U	
3.3	335				U		
4.7	475			U			
6.8	685						
10	106	U			I[(M)]	H/V	
15	156				H	V	
22	226				H		
33	336			H			
47	476		H			T	
68	686				T		
100	107				T		

Released ratings (M tolerance only)
Note: Voltage ratings are minimum values. KYOCERA AVX reserves the right to supply higher voltage ratings in the same case size, to the same reliability standards.

The Important Information/Disclaimer is incorporated in the catalog where these specifications came from or available online at www.kyocera-avx.com/disclaimer/ by reference and should be reviewed in full before placing any order.

RATINGS & PART NUMBER REFERENCE

Part Number	Case Size	Capacitance (µF)	Rated Voltage (V)	Rated Temp. (°C)	Category Voltage (V)	Category Temperature (°C)	DCL Max. (µA)	DF Max. (%)	ESR Max. @ 100kHz (Ω)	100kHz RMS Current (mA)			Product Category	MSL
										25°C	85°C	125°C		
2 Volt														
TACK335*002#TA	K	3.3	2	85	1.3	125	0.5	8	15	32	28	13	3	1
TACL335*002#TA	L	3.3	2	85	1.3	125	0.5	6	7.5	58	52	23	2	1
TACK475*002#TA	K	4.7	2	85	1.3	125	0.5	12	15	32	28	13	3	1
TACL475*002#TA	L	4.7	2	85	1.3	125	0.5	6	7.5	58	52	23	1	1
TACK685*002#TA	K	6.8	2	85	1.3	125	0.5	20	15	32	28	13	3	1
TACL685*002#TA	L	6.8	2	85	1.3	125	0.5	6	7.5	58	52	23	2	1
TACK106*002#TA	K	10	2	85	1.3	125	0.5	15	15	32	28	13	3	1
TACL106*002#TA	L	10	2	85	1.3	125	0.5	10	7.5	58	52	23	3	1
TACU106*002#TA	U	10	2	85	1.3	125	0.5	8	5	84	75	33	1	1
TACR226*002#TA	R	22	2	85	1.3	125	0.5	8	5	95	85	38	1	1
TACR336*002#TA	R	33	2	85	1.3	125	0.7	10	5	95	85	38	2	1
TACR476*002#TA	R	47	2	85	1.3	125	0.9	10	5	95	85	38	2	1
TACR686*002#TA	R	68	2	85	1.3	125	1.4	14	5	95	85	38	2	1
TACA157*002#TA	A	150	2	85	1.3	125	3	20	1	200	180	80	2	1
3 Volt														
TACK225*003#TA	K	2.2	3	85	2	125	0.5	6	15	32	28	13	2	1
TACL225*003#TA	L	2.2	3	85	2	125	0.5	6	7.5	58	52	23	1	1
TACK335*003#TA	K	3.3	3	85	2	125	0.5	8	15	32	28	13	3	1
TACL335*003#TA	L	3.3	3	85	2	125	0.5	6	7.5	58	52	23	2	1
TACK475*003#TA	K	4.7	3	85	2	125	0.5	12	15	32	28	13	3	1
TACL475*003#TA	L	4.7	3	85	2	125	0.5	6	7.5	58	52	23	1	1
TACL685*003#TA	L	6.8	3	85	2	125	0.5	6	7.5	58	52	23	2	1
TACL106*003#TA	L	10	3	85	2	125	0.5	10	7.5	58	52	23	3	1
TACR156*003#TA	R	15	3	85	2	125	0.5	8	5	95	85	38	1	1
TACL226*003#TA	L	22	3	85	2	125	0.7	20	7.5	58	52	23	3	1
TACR226*003#TA	R	22	3	85	2	125	0.7	8	5	95	85	38	1	1
TACR336*003#TA	R	33	3	85	2	125	1	10	5	95	85	38	2	1
TACH476*003#TA	H	47	3	85	2	125	1.4	20	5	89	80	36	3	1
TACR476*003#TA	R	47	3	85	2	125	1.5	10	5	95	85	38	2	1
TACA686*003#TA	A	68	3	85	2	125	2	15	2	141	127	57	1	1
TACR686*003#TA	R	68	3	85	2	125	2	14	5	95	85	38	3	1
TACA107*003#TA	A	100	3	85	2	125	3	15	1	200	180	80	2	1
TACR107*003#TA	R	100	3	85	2	125	3	30	5	95	85	38	3	1
4 Volt														
TACL155*004#TA	L	1.5	4	85	2.7	125	0.5	6	7.5	58	52	23	1	1
TACL225*004#TA	L	2.2	4	85	2.7	125	0.5	6	7.5	58	52	23	1	1
TACL335*004#TA	L	3.3	4	85	2.7	125	0.5	6	7.5	58	52	23	2	1
TACL475*004#TA	L	4.7	4	85	2.7	125	0.5	6	7.5	58	52	23	1	1
TACU475*004#TA	U	4.7	4	85	2.7	125	0.5	8	5	84	75	33	1	1
TACL685*004#TA	L	6.8	4	85	2.7	125	0.5	8	7.5	58	52	23	2	1
TACL106*004#TA	L	10	4	85	2.7	125	0.5	10	7.5	58	52	23	2	1
TACR106*004#TA	R	10	4	85	2.7	125	0.5	8	5	95	85	38	1	1
TACL156*004#TA	L	15	4	85	2.7	125	0.6	20	7.5	58	52	23	3	1
TACR156*004#TA	R	15	4	85	2.7	125	0.6	8	5	95	85	38	1	1
TACL226*004#TA	L	22	4	85	2.7	125	0.9	20	7.5	58	52	23	3	1
TACR226*004#TA	R	22	4	85	2.7	125	0.9	8	5	95	85	38	1	1
TACH336*004#TA	H	33	4	85	2.7	125	1.3	14	5	89	80	36	2	1
TACR336*004#TA	R	33	4	85	2.7	125	1.3	10	5	95	85	38	2	1
TACR476*004#TA	R	47	4	85	2.7	125	1.9	14	5	95	85	38	3	1
TACA686*004#TA	A	68	4	85	2.7	125	2.7	15	1	200	180	80	1	1
TACA107*004#TA	A	100	4	85	2.7	125	4	20	1	200	180	80	2	1
TACR107*004#TA	R	100	4	85	2.7	125	4	30	5	95	85	38	3	1
6.3 Volt														
TACK105*006#TA	K	1	6.3	85	4	125	0.5	6	15	32	28	13	2	1
TACL105*006#TA	L	1	6.3	85	4	125	0.5	6	7.5	58	52	23	1	1
TACL155*006#TA	L	1.5	6.3	85	4	125	0.5	6	7.5	58	52	23	1	1
TACK225*006#TA	K	2.2	6.3	85	4	125	0.5	8	15	32	28	13	3	1
TACL225*006#TA	L	2.2	6.3	85	4	125	0.5	6	7.5	58	52	23	1	1
TACL335*006#TA	L	3.3	6.3	85	4	125	0.5	6	7.5	58	52	23	2	1
TACU335*006#TA	U	3.3	6.3	85	4	125	0.5	8	5	84	75	33	1	1
TACL475*006#TA	L	4.7	6.3	85	4	125	0.5	8	7.5	58	52	23	2	1
TACL685*006#TA	L	6.8	6.3	85	4	125	0.5	10	7.5	58	52	23	2	1
TACR685*006#TA	R	6.8	6.3	85	4	125	0.5	8	5	95	85	38	1	1
TACI106M006#TA	I	10	6.3	85	4	125	0.6	20	5	84	75	33	2	1
TACL106*006#TA	L	10	6.3	85	4	125	0.6	10	6	65	58	26	2	1
TACR106*006#TA	R	10	6.3	85	4	125	0.6	8	5	95	85	38	1	1
TACH156*006#TA	H	15	6.3	85	4	125	0.9	8	5	89	80	36	3	1
TACL156*006#TA	L	15	6.3	85	4	125	0.9	20	7.5	58	52	23	3	1
TACR156*006#TA	R	15	6.3	85	4	125	0.9	8	5	95	85	38	1	1

RATINGS & PART NUMBER REFERENCE

Part Number	Case Size	Capacitance (µF)	Rated Voltage (V)	Rated Temp. (°C)	Category Voltage (V)	Category Temperature (°C)	DCL Max. (µA)	DF Max. (%)	ESR Max. @ 100kHz (Ω)	100kHz RMS Current (mA)			Product Category	MSL
										25°C	85°C	125°C		
TACH226*006#TA	H	22	6.3	85	4	125	1.4	10	5	89	80	36	2	1
TACR226*006#TA	R	22	6.3	85	4	125	1.4	10	5	95	85	38	1	1
TACR336*006#TA	R	33	6.3	85	4	125	2.1	12	5	95	85	38	2	1
TACA476*006#TA	A	47	6.3	85	4	125	3	15	1	200	180	80	1	1
TACR476*006#TA	R	47	6.3	85	4	125	3	20	5	95	85	38	3	1
TACT686*006#TA	T	68	6.3	85	4	125	4.3	15	1	200	180	80	2	1
TACA107*006#TA	A	100	6.3	85	4	125	6.3	20	1	200	180	80	2	1
TACT107*006#TA	T	100	6.3	85	4	125	6.3	12	1	200	180	80	2	1
10 Volt														
TACK154*010#TA	K	0.15	10	85	7	125	0.5	6	40	19	17	8	1	1
TACK224*010#TA	K	0.22	10	85	7	125	0.5	6	30	22	20	9	1	1
TACK334*010#TA	K	0.33	10	85	7	125	0.5	6	20	27	25	11	1	1
TACK474*010#TA	K	0.47	10	85	7	125	0.5	6	15	32	28	13	1	1
TACL474*010#TA	L	0.47	10	85	7	125	0.5	6	7.5	58	52	23	1	1
TACK684*010#TA	K	0.68	10	85	7	125	0.5	8	15	32	28	13	2	1
TACL684*010#TA	L	0.68	10	85	7	125	0.5	6	7.5	58	52	23	1	1
TACK105*010#TA	K	1	10	85	7	125	0.5	6	15	32	28	13	2	1
TACL105*010#TA	L	1	10	85	7	125	0.5	6	7.5	58	52	23	1	1
TACR105*010#TA	R	1	10	85	7	125	0.5	6	7	80	72	32	1	1
TACL155*010#TA	L	1.5	10	85	7	125	0.5	6	7.5	58	52	23	1	1
TACL225*010#TA	L	2.2	10	85	7	125	0.5	6	7.5	58	52	23	1	1
TACU225*010#TA	U	2.2	10	85	7	125	0.5	8	5	84	75	33	1	1
TACL335*010#TA	L	3.3	10	85	7	125	0.5	8	7.5	58	52	23	2	1
TACR335*010#TA	R	3.3	10	85	7	125	0.5	8	5	95	85	38	1	1
TACL475*010#TA	L	4.7	10	85	7	125	0.5	10	6	65	58	26	2	1
TACR475*010#TA	R	4.7	10	85	7	125	0.5	8	6	87	78	35	1	1
TACL685*010#TA	L	6.8	10	85	7	125	0.7	20	7,5	58	52	23	3	1
TACR685*010#TA	R	6.8	10	85	7	125	0.7	8	5	95	85	38	1	1
TACH106*010#TA	H	10	10	85	7	125	1.0	8	5	89	80	36	2	1
TACL106*010#TA	L	10	10	85	7	125	1	20	7.5	58	52	23	3	1
TACR106*010#TA	R	10	10	85	7	125	1	8	5	95	85	38	1	1
TACV106*010#TA	V	10	10	85	7	125	1.0	10	2	132	119	53	2	1
TACR156*010#TA	R	15	10	85	7	125	1.5	10	5	95	85	38	1	1
TACV156*010#TA	V	15	10	85	7	125	1.5	10	2	132	119	53	2	1
TACR226*010#TA	R	22	10	85	7	125	2.2	14	5	95	85	38	2	1
TACA336*010#TA	A	33	10	85	7	125	3.3	12	1	200	180	80	1	1
TACR336*010#TA	R	33	10	85	7	125	3.3	20	5	95	85	38	3	1
TACB476*010#TA	B	47	10	85	7	125	4.7	15	1	200	180	80	1	1
TACT476*010#TA	T	47	10	85	7	125	4.7	12	1	200	180	80	1	1
16 Volt														
TACK104*016#TA	K	0.1	16	85	10	125	0.5	6	40	19	17	8	1	1
TACK154*016#TA	K	0.15	16	85	10	125	0.5	6	30	22	20	9	1	1
TACK224*016#TA	K	0.22	16	85	10	125	0.5	6	20	27	25	11	1	1
TACK334*016#TA	K	0.33	16	85	10	125	0.5	6	20	27	25	11	1	1
TACL474*016#TA	L	0.47	16	85	10	125	0.5	6	7.5	58	52	23	1	1
TACL684*016#TA	L	0.68	16	85	10	125	0.5	6	7.5	58	52	23	1	1
TACL105*016#TA	L	1	16	85	10	125	0.5	6	7.5	58	52	23	1	1
TACU105*016#TA	U	1	16	85	10	125	0.5	8	5	84	75	33	1	1
TACL225*016#TA	L	2.2	16	85	10	125	0.5	10	7.5	58	52	23	1	1
TACR106*016#TA	R	10	16	85	10	125	1.6	10	5	95	85	38	2	1
20 Volt														
TACK224*020#TA	K	0.22	20	85	13	125	0.5	6	20	27	25	11	1	1
TACR335*020#TA	R	3.3	20	85	13	125	0.7	8	5	95	85	38	1	1
TACR475*020#TA	R	4.7	20	85	13	125	0.9	8	5	95	85	38	1	1
25 Volt														
TACR105*025#TA	R	1	25	85	17	125	0.5	8	5	95	85	38	1	1

Moisture Sensitivity Level (MSL) is defined according to J-STD-020. All technical data relates to an ambient temperature of +25°C. Capacitance and DF are measured at 120Hz, 0.5V RMS with a maximum DC bias of 2.2 volts. DCL is measured at rated voltage after 5 minutes. For typical weight and composition, refer to the *Product Safety Information Datasheet* at the end of the Polymer, Tantalum and Niobium Oxide Capacitors catalog.

NOTE: KYOCERA AVX reserves the right to supply higher voltage ratings or tighter tolerance part in the same case size, to the same reliability standards.

QUALIFICATION TABLE – CATEGORY 1

TEST	TAC series (Temperature range -55°C to +125°C)											
	Condition				Characteristics							
Endurance	Apply rated voltage (Ur) at 85°C and / or category voltage (Uc) at 125°C for 2000 hours through a circuit impedance of ≤0.1Ω/V.			Visual examination	no visible damage							
				DCL	1.25 x initial limit							
				ΔC/C	within ±10% of initial value							
				DF	1.5 x initial limit							
				ESR	1.5 x initial limit							
Humidity	Store at 40°C and 90-95% relative humidity for 1344 hours, with no applied voltage.			Visual examination	no visible damage							
				DCL	initial limit							
				ΔC/C	within ±5% of initial value							
				DF	1.2 x initial limit							
				ESR	1.2 x initial limit							
Temperature Stability	Step	Temperature°C	Duration(min)		+20°C	-55°C	+20°C	+85°C	+125°C	+20°C		
	1	+20	15	DCL	IL*	n/a	IL*	10 x IL*	12.5 x IL*	IL*		
	2	-55	15	ΔC/C	n/a	+0/-10%	±5%	+10/-0%	+15/-0%	±5%		
	3	+20	15	DF	IL*	1.5x IL*	IL*	1.5x IL*	2x IL*	IL*		
	4	+85	15	ESR	IL*	1.25 x IL*	IL*	1.25 x IL*	2 x IL*	IL*		
	5	+125	15									
	6	+20	15									
Surge Voltage	Apply 1.3x rated voltage (Ur) at 85°C for 1000 cycles of duration 6 min (30 sec charge, 5 min 30 sec discharge) through a charge / discharge resistance of 1000Ω			Visual examination	no visible damage							
				DCL	initial limit							
				ΔC/C	within ±10% of initial value							
				DF	initial limit							
				ESR	initial limit							

*Initial Limit

QUALIFICATION TABLE – CATEGORY 2

TEST	TAC series (Temperature range -55°C to +125°C)											
	Condition				Characteristics							
Endurance	Apply rated voltage (Ur) at 85°C and / or category voltage (Uc) at 125°C for 2000 hours through a circuit impedance of ≤0.1Ω/V.			Visual examination	no visible damage							
				DCL	1.25 x initial limit							
				ΔC/C	within ±15% of initial value							
				DF	1.5 x initial limit							
				ESR	1.5 x initial limit							
Humidity	Store at 40°C and 90-95% relative humidity for 1344 hours, with no applied voltage.			Visual examination	no visible damage							
				DCL	initial limit							
				ΔC/C	within ±10% of initial value							
				DF	1.2 x initial limit							
				ESR	1.2 x initial limit							
Temperature Stability	Step	Temperature°C	Duration(min)		+20°C	-55°C	+20°C	+85°C	+125°C	+20°C		
	1	+20	15	DCL	IL*	n/a	IL*	10 x IL*	12.5 x IL*	IL*		
	2	-55	15	ΔC/C	n/a	+0/-15%	±5%	+15/-0%	+20/-0%	±5%		
	3	+20	15	DF	IL*	1.5x IL*	IL*	1.5x IL*	2x IL*	IL*		
	4	+85	15	ESR	IL*	1.25 x IL*	IL*	1.25 x IL*	2 x IL*	IL*		
	5	+125	15									
	6	+20	15									
Surge Voltage	Apply 1.3x rated voltage (Ur) at 85°C for 1000 cycles of duration 6 min (30 sec charge, 5 min 30 sec discharge) through a charge / discharge resistance of 1000Ω			Visual examination	no visible damage							
				DCL	1.5 x initial limit							
				ΔC/C	within ±15% of initial value							
				DF	1.5 x initial limit							
				ESR	1.5 x initial limit							

*Initial Limit

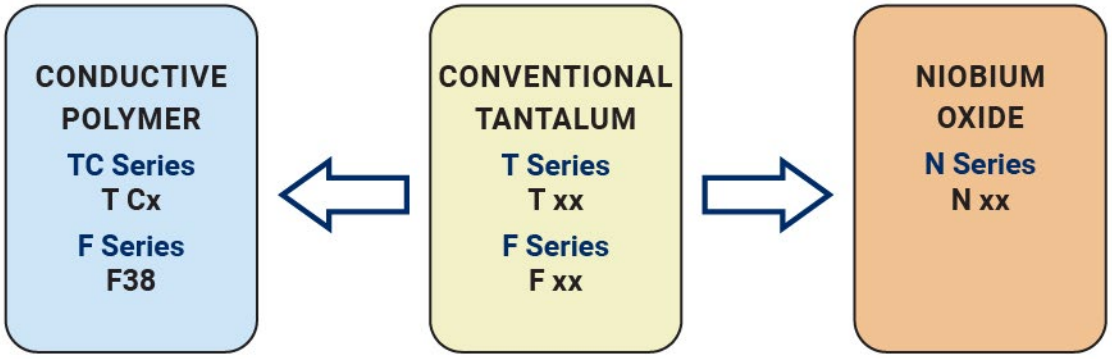
QUALIFICATION TABLE – CATEGORY 3

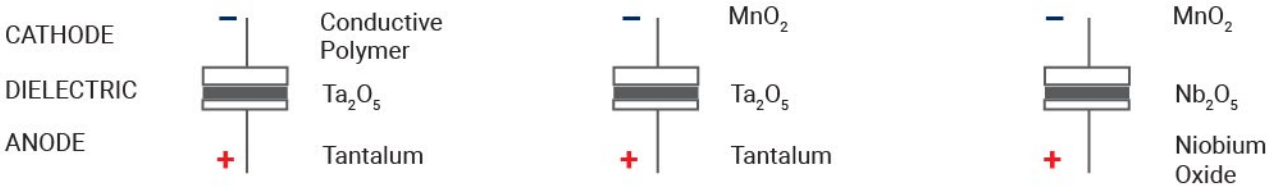
<table>
<tr><th rowspan="2">TEST</th><th colspan="9">TAC series (Temperature range -55°C to +125°C)</th></tr>
<tr><th colspan="4">Condition</th><th></th><th colspan="5">Characteristics</th></tr>
<tr><td rowspan="5">Endurance</td><td colspan="4" rowspan="5">Apply rated voltage (Ur) at 85ºC and / or category voltage (Uc) at 125ºC for 2000 hours through a circuit impedance of ≤0.1Ω/V.</td><td>Visual examination</td><td colspan="5">no visible damage</td></tr>
<tr><td>DCL</td><td colspan="5">1.25 x initial limit</td></tr>
<tr><td>ΔC/C</td><td colspan="5">within ±30% of initial value</td></tr>
<tr><td>DF</td><td colspan="5">1.5 x initial limit</td></tr>
<tr><td>ESR</td><td colspan="5">1.5 x initial limit</td></tr>
<tr><td rowspan="5">Humidity</td><td colspan="4" rowspan="5">Store at 40ºC and 90-95% relative humidity for 1344 hours, with no applied voltage.</td><td>Visual examination</td><td colspan="5">no visible damage</td></tr>
<tr><td>DCL</td><td colspan="5">2 x initial limit</td></tr>
<tr><td>ΔC/C</td><td colspan="5">within ±30% of initial value</td></tr>
<tr><td>DF</td><td colspan="5">1.5 x initial limit</td></tr>
<tr><td>ESR</td><td colspan="5">1.25 x initial limit</td></tr>
<tr><td rowspan="8">Temperature Stability</td><td>Step</td><td colspan="2">Temperature°C</td><td>Duration(min)</td><td></td><td>+20ºC</td><td>-55ºC</td><td>+20ºC</td><td>+85ºC</td><td>+125ºC</td><td>+20ºC</td></tr>
<tr><td>1</td><td colspan="2">+20</td><td>15</td><td>DCL</td><td>IL*</td><td>n/a</td><td>IL*</td><td>10 x IL*</td><td>12.5 x IL*</td><td>IL*</td></tr>
<tr><td>2</td><td colspan="2">-55</td><td>15</td><td>ΔC/C</td><td>n/a</td><td>+0/-25%</td><td>±5%</td><td>+20/-0%</td><td>+25/-0%</td><td>±20%</td></tr>
<tr><td>3</td><td colspan="2">+20</td><td>15</td><td>DF</td><td>IL*</td><td>1.5xIL*</td><td>IL*</td><td>1.5xIL*</td><td>2xIL*</td><td>1.5xIL*</td></tr>
<tr><td>4</td><td colspan="2">+85</td><td>15</td><td>ESR</td><td>IL*</td><td>1.25 x IL*</td><td>IL*</td><td>1.25 x IL*</td><td>2 x IL*</td><td>1.5 x IL*</td></tr>
<tr><td>5</td><td colspan="2">+125</td><td>15</td><td colspan="7"></td></tr>
<tr><td>6</td><td colspan="2">+20</td><td>15</td><td colspan="7"></td></tr>
<tr><td rowspan="5">Surge Voltage</td><td colspan="4" rowspan="5">Apply 1.3x rated voltage (Ur) at 85°C for 1000 cycles of duration 6 min (30 sec charge, 5 min 30 sec discharge) through a charge / discharge resistance of 1000Ω</td><td>Visual examination</td><td colspan="5">no visible damage</td></tr>
<tr><td>DCL</td><td colspan="5">2 x initial limit</td></tr>
<tr><td>ΔC/C</td><td colspan="5">within ±30% of initial value</td></tr>
<tr><td>DF</td><td colspan="5">2 x initial limit</td></tr>
<tr><td>ESR</td><td colspan="5">2 x initial limit</td></tr>
</table>

*Initial Limit

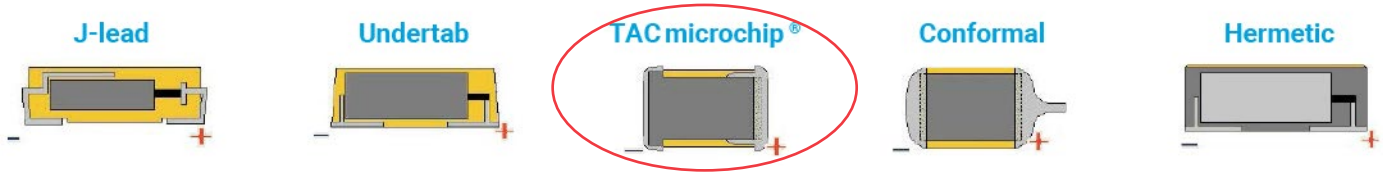
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SOLID ELECTROLYTIC CAPACITOR ROADMAP





FIVE CAPACITOR CONSTRUCTION STYLES



SERIES LINE UP : Conventional SMD MnO_2

